

PROBLEM SET 2—SOLUTIONS

ECON301—ECONOMETRICS I

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These are my suggested solutions. It is your responsibility to cross-check this document with your own answers.

1. (a) $E(\bar{Y}|X's) = E\left[\frac{1}{n} \sum Y_i|X's\right] = \frac{1}{n} \sum E[Y_i|X's] = \frac{1}{n} \sum E[Y_i|X_i] = \frac{1}{n} \sum (\beta_0 + \beta_1 X_i) = \beta_0 + \beta_1 \bar{X}$. The third equality is by strong exogeneity and the fourth equality is by assumption 2 (linearity).
 - (b) $E(\hat{\beta}_0 + \hat{\beta}_1 X_1|X's) = E(\hat{\beta}_0|X's) + X_1 E(\hat{\beta}_1|X's) = \beta_0 + \beta_1 X_1$, where the last equality uses the results from theorem 3.3.
 - (c) $\text{Var}(\hat{\beta}_0 + \hat{\beta}_1 X_1|X's) = \text{Var}(\hat{\beta}_0|X's) + X_1^2 \text{Var}(\hat{\beta}_1|X's) + 2X_1 \text{Cov}(\hat{\beta}_0, \hat{\beta}_1) = \frac{\sigma^2}{s_{xx}n} \sum_i X_i^2 + X_1^2 \frac{\sigma^2}{s_{xx}} - 2X_1 \frac{\sigma^2}{s_{xx}} \bar{X} = \frac{\sigma^2}{s_{xx}} \left[(1/n) \sum_i X_i^2 + X_1^2 - 2X_1 \bar{X} \right]$. The first equality is by theorem 2.3 and 2.4. The rest is by theorem 3.3.
 - (d) $\text{Var}(\bar{Y} - \beta_1 \bar{X}|X's) = \text{Var}(\bar{Y}|X's) = \frac{1}{n^2} \sum_i \text{Var}(Y_i|X's) = \frac{1}{n^2} n \sigma^2 = \frac{\sigma^2}{n}$. The first equality is because the term $\beta_1 \bar{X}$ is a constant conditional on the $X's$. The second equality is the variance calculation formula from theorem 2.2 and also uses strong exogeneity.¹ The third equality is by assumption 4 of homoskedasticity.
 - (e) $\text{Cov}(\bar{Y}, \hat{\beta}_1|X's) = \frac{1}{ns_{xx}} \text{Cov}(\sum_i Y_i, s_{xy}|X's) = \frac{1}{ns_{xx}} \text{Cov}(\sum_i Y_i, \sum_i (X_i - \bar{X})Y_i|X's) = \frac{1}{ns_{xx}} \sum_i (X_i - \bar{X}) \text{Cov}(Y_i, Y_i|X's) = \frac{1}{ns_{xx}} \sum_i (X_i - \bar{X}) \text{Var}(Y_i|X's) = \frac{1}{ns_{xx}} \sum_i (X_i - \bar{X}) \sigma^2 = \frac{\sigma^2}{ns_{xx}} \sum_i (X_i - \bar{X}) = 0$. Here, most of the computations use the results from theorem 2.3. The third equality uses strong exogeneity.²
 - (f) $E(\bar{Y} - \hat{\beta}_1 X_1|X's) = E(\bar{Y}|X's) - X_1 E(\hat{\beta}_1|X's) = \beta_0 + \beta_1 \bar{X} - X_1 \beta_1 = \beta_0 + \beta_1 (\bar{X} - X_1)$. The first equality is by linearity of conditional expectation and the second equality is from task 1.a in this problem set and theorem 3.3, i.e., the conditional unbiasedness of the LSE.
 - (g) $\text{Var}(\bar{Y} - \hat{\beta}_1 X_1|X's) = \text{Var}(\bar{Y}|X's) + X_1^2 \text{Var}(\hat{\beta}_1|X's) - 2X_1 \text{Cov}(\bar{Y}, \hat{\beta}_1) = \frac{\sigma^2}{n} + \frac{\sigma^2}{s_{xx}} X_1^2 - \frac{\sigma^2}{s_{xx}} \left(\frac{s_{xx}}{n} + X_1^2 \right) = \frac{\sigma^2}{s_{xx}} \left(\frac{1}{n} \sum_i X_i^2 - \bar{X}^2 + X_1^2 \right)$. The first equality is the variance calculation formula. The second equality uses the results from items 1.d, 1.e, and theorem 3.3. The rest is just algebra and the result from lemma 3.1.
2. $E(\hat{\beta}_0|X's) = E(\bar{Y} - \hat{\beta}_1 \bar{X}|X's) = E(\bar{Y}|X's) - \bar{X} E(\hat{\beta}_1|X's) = \beta_0 + \beta_1 \bar{X} - \bar{X} \beta_1 = \beta_0$. The results follow from the results in item 1.a in this problem set and the hint, which one of the results proven in theorem 3.3.
 3. In Lecture 3, part 3, we calculated the expression for $\text{Var}(\hat{\beta}_1|X's)$. We need to calculate the remaining expressions in the variance-covariance matrix: $\text{Var}(\hat{\beta}_0|X's)$ and $\text{Cov}(\hat{\beta}_0, \hat{\beta}_1|X's)$. We start by calculating the conditional variance of $\hat{\beta}_0$:

$$\begin{aligned} \text{Var}(\hat{\beta}_0|X's) &= \text{Var}(\bar{Y} - \hat{\beta}_1 \bar{X}|X's) = \text{Var}(\bar{Y}|X's) + \bar{X}^2 \text{Var}(\hat{\beta}_1|X's) - 2\bar{X} \text{Cov}(\bar{Y}, \hat{\beta}_1) \\ &= \frac{\sigma^2}{n} + \frac{\sigma^2}{s_{xx} \bar{X}^2} - 0 = \frac{\sigma^2}{s_{xx}} \left(\frac{s_{xx}}{n} + \bar{X}^2 \right) = \frac{\sigma^2}{s_{xx}} \left(\frac{1}{n} \sum_i X_i^2 - \bar{X}^2 + \bar{X}^2 \right) \\ &= \frac{\sigma^2}{s_{xx}} \left(\frac{1}{n} \sum_i X_i^2 \right) \end{aligned}$$

The calculation above uses many of the results from task 1 in this problem set. The third equality uses 1.d and 1.e, as well as the expression for the variance of $\hat{\beta}_1$ that was calculated in theorem 3.3; it also uses item 1.f as inspiration about how to compute this variance. The penultimate equality uses the algebraic result from lemma 3.1.

Next, we calculate the conditional covariance between $\hat{\beta}_0$ and $\hat{\beta}_1$:

$$\begin{aligned} \text{Cov}(\hat{\beta}_0, \hat{\beta}_1|X's) &= \text{Cov}(\bar{Y} - \hat{\beta}_1 \bar{X}, \hat{\beta}_1|X's) = \text{Cov}(\bar{Y}, \hat{\beta}_1|X's) - \bar{X} \text{Cov}(\hat{\beta}_1, \hat{\beta}_1|X's) = -\bar{X} \text{Var}(\hat{\beta}_1|X's) \\ &= -\frac{\sigma^2}{s_{xx}} \bar{X} \end{aligned}$$

¹An attentive student should recognize that in the variance computation, a bunch of covariance terms between Y_i and Y_j for $i \neq j$ should appear. I skipped them because by strong exogeneity, they equal zero. For a formal proof of this fact, see exercise 5d in this problem set.

²Once again, you should recognize that in the sum of covariances computation, a bunch of covariance terms between Y_i and Y_j for $i \neq j$ should appear. I skipped them because by strong exogeneity, they equal zero. For a formal proof of this fact, see exercise 5d in this problem set.

In this calculation, we exploit all the results from task 1 in this problem set. The penultimate inequality is due to task 1.e. The last equality exploits our knowledge of $\text{Var}(\hat{\beta}_1|X's)$, which we derived in the proof of theorem 3.3.

Note that the results from task 2 and 3 in this problem set close the remaining gaps in the proof of theorem 3.3 which I gave in the lecture.

4. First note that the population parameters are: $\beta_0 = 3$, $\beta_1 = 2$ and $\sigma^2 = 3$.

- (a) With the data that we have, we calculate $\bar{X} = 3.5$. Now we can calculate $s_{xx} = \sum_{i=1}^3 X_i^2 - 3(3.6^2) = 15.5$. Since $s_{xx} > 0$ then assumption 3 is satisfied.
- (b) From item 1.f above we know that $E(\bar{Y} - \hat{\beta}_1 X_1|X's) = \beta_0 + \beta_1(\bar{X} - X_1) = 3 + 2(3.5 - 1) = 8$.
Now we calculate the expression for $\text{Var}(\hat{\beta}_0 + \hat{\beta}_1|X's) = \text{Var}(\hat{\beta}_0|X's) + \text{Var}(\hat{\beta}_1|X's) + 2\text{Cov}(\hat{\beta}_0, \hat{\beta}_1) = \frac{\sigma^2}{s_{xx}} \left(\frac{1}{n} \sum_i X_i^2 + 1 - 2\bar{X} \right)$. For the data given: $\text{Var}(\hat{\beta}_0 + \hat{\beta}_1|X's) = \frac{3}{15.5} \left(\frac{1}{3}(52.25) + 1 - 2(3.5) \right) = 2.209677$
- (c) From the expression in theorem 3.3 and using the data from this problem, we have that

$$\text{Var}(\underline{\hat{\beta}}|X's) = \frac{3}{15.5} \begin{bmatrix} 52.25/3 & -3.5 \\ -3.5 & 1 \end{bmatrix} = \begin{bmatrix} 3.3710 & -0.6774 \\ -0.6774 & 0.1935 \end{bmatrix}$$

5. Because of independence between each of the $(X_i, Y_i)_{i=1}^n$, we have that

$$f(x_1, y_1, \dots, x_n, y_n) = \prod_{i=1}^n f(x_i, y_i).$$

(a) We have that

$$f(y_1, \dots, y_n | x_1, \dots, x_n) = \frac{f(x_1, y_1, \dots, x_n, y_n)}{f(x_1, \dots, x_n)} = \frac{\prod_{i=1}^n f(x_i, y_i)}{f(x_1, \dots, x_n)}. \quad (1)$$

The denominator is equal to

$$\begin{aligned} f(x_1, \dots, x_n) &= \int \cdots \int f(x_1, y_1, \dots, x_n, y_n) dy_1 \cdots dy_n = \int \cdots \int \prod_{i=1}^n f(x_i, y_i) dy_1 \cdots dy_n \\ &= \prod_{i=1}^n \left(\int f(x_i, y_i) dy_i \right) = \prod_{i=1}^n f(x_i) \end{aligned} \quad (2)$$

Combining (1) and (2) we get

$$f(y_1, \dots, y_n | x_1, \dots, x_n) = \frac{\prod_{i=1}^n f(x_i, y_i)}{\prod_{i=1}^n f(x_i)} = \prod_{i=1}^n \frac{f(x_i, y_i)}{f(x_i)} = \prod_{i=1}^n f(y_i | x_i).$$

- (b) For simplicity, we will introduce the following notation: for any vector with n components $(z_1, \dots, z_n)$, let $z_{-i} = (z_j)_{j \neq i}$, i.e., the vector with $n - 1$ components in which we delete the component i but all other ones remain. Similarly, denote by $dz_{-i} = \times_{j \neq i} dz_j$, i.e., the product of all differentials dz_j except dz_i . Then we have that

$$\begin{aligned} f(y_i | x_1, \dots, x_n) &= \int \cdots \int f(y_i, y_{-i} | x_1, \dots, x_n) dy_{-i} = \int \cdots \int \prod_{j=1}^n f(y_j | x_j) dy_{-i} \\ &= \int \cdots \int f(y_i | x_i) \prod_{j \neq i} f(y_j | x_j) dy_{-i} = f(y_i | x_i) \prod_{j \neq i} \left(\int f(y_j | x_j) dy_j \right) \\ &= f(y_i | x_i), \end{aligned}$$

because all the integrands $f(y_j | x_j)$, $j = 1, \dots, n$, $j \neq i$ integrate to one.

- (c) Similarly as above, for any vector with n components $(z_1, \dots, z_n)$, let $z_{-i, -j} = (z_k)_{k \neq i, k \neq j}$, i.e., the vector with $n - 2$ components in which we delete the i th and j th entries but all other ones remain. Similarly, denote by $dz_{-i, -j} = \times_{k \neq i, k \neq j} dz_k$, i.e., the product of all differentials dz_k except dz_i and dz_j . Then we

have that

$$\begin{aligned}
f(y_i, y_j | x_1, \dots, x_n) &= \int \cdots \int f(y_i, y_j, y_{-i,-j} | x_1, \dots, x_n) dy_{-i,-j} = \int \cdots \int \prod_{k=1}^n f(y_k | x_k) dy_{-i,-j} \\
&= \int \cdots \int f(y_i | x_i) f(y_j | x_j) \prod_{k \neq i, k \neq j} f(y_k | x_k) dy_{-i,-j} \\
&= f(y_i | x_i) f(y_j | x_j) \prod_{k \neq i, k \neq j} \left(\int f(y_k | x_k) dy_k \right) \\
&= f(y_i | x_i) f(y_j | x_j),
\end{aligned}$$

because all the integrands $f(y_k | x_k), k = 1, \dots, n, k \neq i, k \neq j$ integrate to one.

- (d) First note that since we have independent and identically distributed random variables, $\text{Var}(Y_1 | X_1) = \text{Var}(Y_i | X_i)$ for $i = 1, \dots, n$. With this in mind, we are going to calculate the covariance next. We need to analyze two cases:

Case 1: $i \neq j$ When the two terms are different we have that:

$$\begin{aligned}
\text{Cov}(Y_i, Y_j | X_1, \dots, X_n) &= E[Y_i Y_j | X_1, \dots, X_n] - E[Y_i | X_1, \dots, X_n] E[Y_j | X_1, \dots, X_n] \\
&= \iint y_i y_j f(y_i, y_j | x_1, \dots, x_n) dy_i dy_j \\
&\quad - \left(\int y_i f(y_i | x_1, \dots, x_n) dy_i \right) \left(\int y_j f(y_j | x_1, \dots, x_n) dy_j \right) \\
&= \iint y_i y_j f(y_i | x_i) f(y_j | x_j) dy_i dy_j - \left(\int y_i f(y_i | x_i) dy_i \right) \left(\int y_j f(y_j | x_j) dy_j \right) \\
&= \left(\int y_i f(y_i | x_i) dy_i \right) \left(\int y_j f(y_j | x_j) dy_j \right) - \left(\int y_i f(y_i | x_i) dy_i \right) \left(\int y_j f(y_j | x_j) dy_j \right) \\
&= 0
\end{aligned} \tag{3}$$

Case 2: $i = j$ For this case we have that

$$\begin{aligned}
\text{Cov}(Y_i, Y_j | X_1, \dots, X_n) &= \text{Cov}(Y_i, Y_i | X_1, \dots, X_n) = \text{Var}(Y_i | X_1, \dots, X_n) \\
&= E[Y_i^2 | X_1, \dots, X_n] - (E[Y_i | X_1, \dots, X_n])^2 \\
&= \int y_i^2 f(y_i | x_1, \dots, x_n) dy_i - \left(\int y_i f(y_i | x_1, \dots, x_n) dy_i \right)^2 \\
&= \int y_i^2 f(y_i | x_i) dy_i - \left(\int y_i f(y_i | x_i) dy_i \right)^2 \\
&= E[Y_i^2 | X_i] - (E[Y_i | X_i])^2 \\
&= \text{Var}(Y_i | X_i) = \text{Var}(Y_1 | X_1),
\end{aligned} \tag{4}$$

where the fourth equality comes from item (b) in this task and the last equality comes from the fact the random variables are identically distributed, as mentioned earlier. Combining (3) and (4) we get the statement written in the question.

6. You have met this distribution before: in the example from lecture 2 and in problem set 1. First of all note that as we computed in lecture 2: $E[Y | X = x] = \frac{3x+2}{6x+3}$. Thus, assumption 2 from lecture 3 about the linearity of the conditional expectation will not be met. However, assumption 1 is met because we do have a random sample with n independent observations. Thus, strong exogeneity will be true, and we will have that $E[Y_i | X_i] = \frac{3X_i+2}{6X_i+3}$.

Let's compute $E(\hat{\beta}_1 | X's)$:

$$\begin{aligned}
E(\hat{\beta}_1 | X's) &= E(s_{xy} / s_{xx} | X's) = \frac{1}{s_{xx}} E[s_{xy} | X's] = \frac{1}{s_{xx}} E \left[\sum_i (X_i - \bar{X}) Y_i | X's \right] \\
&= \frac{1}{s_{xx}} \sum_i (X_i - \bar{X}) E[Y_i | X's] = \frac{1}{s_{xx}} \sum_i (X_i - \bar{X}) E[Y_i | X_i]
\end{aligned}$$

where the last equality comes from strong exogeneity. Up until now, the calculation looks like the proof of theorem 3.3, but since we do not have linearity, we cannot proceed with the calculation in the same way as we did in the proof of that theorem. However, we do have an expression for the conditional expectation:

$$= \frac{1}{s_{xx}} \sum_i \left\{ (X_i - \bar{X}) \left(\frac{3X_i + 2}{6X_i + 3} \right) \right\}$$

With the data that we have: $E(\hat{\beta}_1|X's) = \frac{-0.11508}{15.5} = -0.00742$.

7. Let me begin by introducing some new notation that will help with the calculations. Denote the components of the variance covariance matrix as follows:

$$\text{Var}(\hat{\beta}|X's) = \sigma^2 \begin{bmatrix} \frac{1}{ns_{xx}} \sum_i X_i^2 & -\frac{\bar{X}}{s_{xx}} \\ -\frac{\bar{X}}{s_{xx}} & \frac{1}{s_{xx}} \end{bmatrix} = \sigma^2 \begin{bmatrix} V_0 & V_{01} \\ V_{01} & V_1 \end{bmatrix}$$

Then note that $\text{Var}(\hat{\beta}_i|X's) = \sigma^2 V_i$. From Theorem 3.4, we know that the LSE has a bivariate normal distribution. It is a well-known result that the marginal distributions from a bivariate normal distribution are also normal. Thus we have that

$$\hat{\beta}_i|X's \sim N(\beta_i, \sigma^2 V_i).$$

Therefore, we have that:

$$\frac{\hat{\beta}_i - \beta_i}{\sigma \sqrt{V_i}} \sim N(0, 1).$$

On the other hand, as explained in the lecture, we have the following:

$$\frac{n-2}{\sigma^2} \hat{\sigma}^2 | X's \sim \chi^2(n-2).$$

Note that $\text{SE}(\hat{\beta}_i|X's) = \hat{\sigma} \sqrt{V_i}$. Thus

$$\frac{\frac{\hat{\beta}_i - \beta_i}{\sigma \sqrt{V_i}}}{\sqrt{\left(\frac{n-2}{\sigma^2} \hat{\sigma}^2\right) / (n-2)}} = \frac{\frac{\hat{\beta}_i - \beta_i}{\sigma \sqrt{V_i}}}{\hat{\sigma} / \sigma} = \frac{\hat{\beta}_i - \beta_i}{\hat{\sigma} \sqrt{V_i}} = \frac{\hat{\beta}_i - \beta_i}{\text{SE}(\hat{\beta}_i|X's)} = t_i$$

Thus, conditional on the $X's$, the t -statistic $t_i \sim t(n-2)$.

8. Using the previous result, a confidence interval for β_1 is $\left[\hat{\beta}_1 \pm t_{\alpha/2}^{(n-2)} \text{SE}(\hat{\beta}_1|X's) \right] = \left[\hat{\beta}_1 \pm t_{\alpha/2}^{(n-2)} \hat{\sigma} \sqrt{V_1} \right]$. For the data that we have, $s_{xx} = 15.5$, so that $V_1 = 1/15.5 = 0.0645$, $\hat{\sigma} = \sqrt{\frac{\sum_i (\hat{\epsilon}_i)^2}{n-2}} = \sqrt{\frac{8.6}{1}} = 2.9326$, and thus $\text{SE}(\hat{\beta}_1|X's) = 2.9396 \sqrt{0.0645} = 0.7449$. For a 95% two-sided confidence interval, we need the critical value for a Student's t distribution with one degree of freedom: $t_{0.025}^{(1)} = 12.7062$. Thus, the confidence interval is $[\hat{\beta}_1 \pm 9.4648]$.